

DO GAME-THEORETIC DECISION RULES IMPROVE TWO-VOICE MUSIC GENERATION?

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ABSTRACT

We ask whether game-theoretic decision rules improve two-voice symbolic music generation beyond a learned joint-compatibility baseline. Using soprano and bass from the Johann Sebastian Bach (JSB) Chorales, we train a multi-head model with voice-specific and pair-compatibility scores. All methods share the same duration-aware model and candidate construction, differing only in pair selection: no-game joint sampling, quantal-response-equilibrium (QRE)-style soft response, Stackelberg leader-follower selection, and pure Nash equilibrium. Across 77 held-out chorales, No-game has the lowest harmonic interval-class and voice-distance Jensen-Shannon (JS) point estimates. QRE shows favorable duration and melodic point estimates but uncertain paired differences; both Stackelberg variants reliably improve local 3-gram diversity matching, while Nash collapses toward sustained, low-diversity output. A blind A/B study with 47 listeners most clearly favors No-game for coherence over Lower Stackelberg. Strategic inference therefore changes generation behavior, but its gains are selective and do not necessarily translate into perceived quality.

1. INTRODUCTION

Polyphonic generation requires both plausible individual voices and coherent vertical interaction. Neural chorale systems such as DeepBach and Coconet model dependencies across simultaneous voices [1, 2], while multi-agent systems explore role-based composition [3]. Game theory offers explicit interaction rules such as soft response, leader-follower decisions, and equilibrium stability [4–6], but whether these rules improve symbolic music generation at inference time remains unclear.

A weak baseline can make strategic inference appear useful simply because it introduces joint modeling. We therefore compare against a strong No-game baseline that already uses a learned upper-lower compatibility score. The central question is whether QRE-, Stackelberg-, or Nash-based selection adds value beyond that learned compatibility. To isolate the effect of the decision rule, all

methods share the same model, state representation, candidate construction, test chorales, and continuation length.

2. METHOD

2.1 Data and State

We use soprano and bass from the voice-separated JSB Chorales dataset [7], with 229/76/77 train/validation/test pieces on a sixteenth-note grid using Musical Instrument Digital Interface (MIDI) pitch and the special tokens REST and HOLD. The state contains a 16-step two-voice context, bar position, and six duration features: hold length, steps since the last onset, and a previous-token-is-HOLD indicator for each voice. These features were added after preliminary experiments showed that stable solvers, especially Stackelberg and Nash, could amplify sustained-note behavior. All methods use the same duration-aware checkpoint and state.

2.2 Shared Utility Model and Decision Rules

A multi-head multilayer perceptron (MLP) outputs upper- and lower-voice private scores and a learned pair-compatibility score:

$$V_U(s, a_U), \quad V_L(s, a_L), \quad C(s, a_U, a_L), \quad (1)$$

where s is the state, a_U, a_L are candidate upper/lower tokens, V_U, V_L are private scores, and C is pair compatibility. Each method forms a 12×12 candidate grid from the top-12 tokens of both private heads.

No-game samples directly from C , testing whether learned joint compatibility is already sufficient. Game-based rules use

$$U_U = C + \alpha V_U, \quad U_L = C + \beta_p V_L, \quad (2)$$

where U_U, U_L are player utilities and α, β_p weight private scores. QRE [4] models simultaneous soft response: each voice probabilistically favors higher-utility actions rather than making a hard best response. It is renormalized over the top- N compatibility-filtered pairs, where N is the number retained; validation selects $\alpha = \beta_p = 0.5$, response softness 0.5, and $N = 20$.

Stackelberg [5] models asymmetric leader-follower interaction. We report the validation-selected lower-leading configuration (Lower STK) and an upper-leading variant



(Upper STK) as a leader ablation. Nash [6] is a strict local-equilibrium ablation: a selected pair must be stable to unilateral deviation, with welfare tie-breaking/fallback. Validation selects $\alpha = \beta_p = 1.0$. QRE, Stackelberg, and Nash are chosen from 9, 6, and 3 validation settings, respectively, whereas No-game is unswept; this asymmetry limits attribution of all performance differences solely to the selection rule.

3. EVALUATION

All methods are evaluated on the same 77 held-out chorales, each seeded by a 16-step real prefix and generating a matched 128-step continuation; the seed is removed before scoring. Hyperparameters are selected on validation data and frozen before testing. We use 10,000 paired bootstrap resamples of the 77 matched continuations to quantify cross-chorale uncertainty while keeping the real test corpus fixed.

Generated and real corpora are compared using lower-is-better JS divergences and scalar gaps. Harmonic interval-class (HIC) JS measures $|s - b| \bmod 12$, where s and b are soprano and bass MIDI pitches; voice-distance (VD) JS retains signed spacing $s - b$. Additional measures include melodic-interval (MI) and pitch-class (PC) JS, hold-ratio, mean VD, pitch-entropy (PE), and unique 3-gram gaps. Figure 1 highlights HIC JS, hold-ratio gap, and unique 3-gram gap as representative measures of vertical coordination, duration behavior, and local diversity. For paired comparisons, $\Delta = \text{metric}_{\text{method}} - \text{metric}_{\text{No-game}}$; negative values favor the compared method, and 95% intervals crossing zero indicate an uncertain difference.

For perceptual evaluation, we recruited an online convenience sample of 47 volunteers through an open call, without screening for musical experience. Of these participants, 23 reported no formal music training, 7 less than one year, 9 one to three years, and 8 more than four years; median self-rated classical-music familiarity was 2 on a 1–5 scale. Participants completed a blind A/B study using validation-selected QRE and Lower STK. The study contained three method pairings with four matched seeds each, for 12 trials total. Each trial presented two 8-bar clips (approximately 21.3 s at 90 BPM) rendered with the same piano timbre and shown as black-screen videos; method

identities were hidden. Participants judged voice coordination, naturalness/coherence, and overall preference, with A, B, or no-clear-preference responses. Responses were aggregated per participant before two-sided exact binomial tests, with Holm correction across nine tests.

4. RESULTS AND DISCUSSION

Figure 1 shows distinct rather than uniformly better strategic behaviors. On HIC JS, QRE is reliably farther from the real corpus, Lower STK also shifts in the unfavorable direction, and Upper STK remains close to No-game. Duration behavior differs: QRE has the lowest hold-ratio-gap point estimate, but its paired interval crosses zero, while both Stackelberg variants have larger gaps. Nash shows by far the largest hold-ratio deviation, consistent with collapse toward excessive sustain.

The clearest positive strategic effect appears in local diversity. Both Lower and Upper STK reliably reduce the unique 3-gram gap relative to No-game, whereas QRE moves in the opposite direction and Nash strongly degrades. Upper STK is particularly informative because its HIC and VD differences from No-game are both small and uncertain while its 3-gram advantage is reliable, illustrating a coordination–diversity trade-off rather than uniform improvement. Secondary metrics support the same interpretation: QRE has the lowest point estimates for both melodic-interval divergences, but their paired intervals include zero; apparent Stackelberg advantages in mean spacing or pitch entropy are likewise uncertain. Pure Nash remains the clearest failure mode, with low diversity and excessive sustain across the full test set.

The listening study gives a complementary perceptual result. Although Lower STK improves local 3-gram matching objectively, listeners favor No-game for perceived coherence (24 vs. 7, 16 ties), the only comparison surviving Holm correction at $p < .05$. No-game also trends favorably against QRE, but those comparisons do not survive correction. Together, the objective and perceptual results show that game-theoretic inference produces repeatable, method-specific changes, yet improvement in an individual corpus statistic does not necessarily imply greater perceived musical quality.

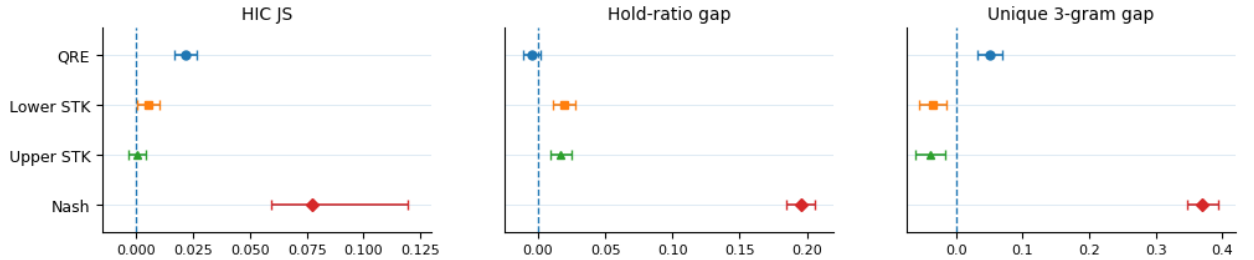


Figure 1. Representative paired-bootstrap effects relative to No-game on the 77 test chorales. Points show method-minus-No-game differences and bars show 95% confidence intervals (CIs). All metrics are lower-is-better; negative values favor the compared method. HIC denotes harmonic interval class.

5. REFERENCES

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